

Polynomials

ISI UGB Mastery

From the Fundamental Theorem of Algebra to Strange Inequalities

The Fundamental Theorem of Algebra

Fundamental Theorem of Algebra.

Every non-constant polynomial $P(x) \in \mathbb{C}[x]$ of degree n has exactly n roots in $\mathbb{C}$, counted with multiplicity.

A historical aside. The popular joke goes:

The Fundamental Theorem of Algebra is neither fundamental, nor a theorem of algebra.

The point of the joke: although the statement is purely algebraic — about roots of polynomials — every known proof relies essentially on *analysis* (continuity, completeness of $\mathbb{R}$ or $\mathbb{C}$). There is no purely algebraic proof.

Yet, for our purposes, it is indispensable: it guarantees that degree controls the number of roots.

The Identity Principle

Corollary (Identity Principle).

If a polynomial of degree at most n vanishes at $n + 1$ distinct points, then it is identically zero.

Equivalently: *if two polynomials of degree $\leq n$ agree at $n + 1$ distinct points, they are equal as polynomials.*

Why it is powerful. It converts a finite verification into a global identity. To prove $P(x) \equiv Q(x)$ for polynomials of degree $\leq n$, it suffices to check equality at any $n + 1$ values you choose — typically the ones that make the algebra trivial.

This single corollary is the engine behind several classical constructions (Lagrange interpolation, finite-difference recursions, etc.).

Example 1 — A Lagrange-style Identity

Problem. Let a, b, c be three distinct real numbers. Define

$$P(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-a)(x-c)}{(b-a)(b-c)} + \frac{(x-a)(x-b)}{(c-a)(c-b)}.$$

Prove that $P(x) \equiv 1$.

Crux. P has degree ≤ 2 . Check P at *three* points $\{a, b, c\}$, then invoke the Identity Principle.

Example 1 — Solution

Solution. Each summand is a quadratic in x , so $\deg P \leq 2$.

Evaluate at $x = a$. The first summand becomes

$$\frac{(a-b)(a-c)}{(a-b)(a-c)} = 1,$$

while the other two summands contain a factor $(x - a)$ and hence vanish. Thus $P(a) = 1$. By symmetry, $P(b) = P(c) = 1$.

Define $Q(x) = P(x) - 1$. Then $\deg Q \leq 2$, and Q vanishes at the three distinct points a, b, c . By the Identity Principle, $Q \equiv 0$, i.e. $P(x) \equiv 1$. ■

Remark. This is the simplest non-trivial case of *Lagrange interpolation*: the unique polynomial of degree $\leq n - 1$ taking prescribed values at n distinct nodes.

Example 2 — A Functional Recursion

Problem. Find all polynomials $P(x)$ satisfying

$$P(x+1) = P(x) + 2x + 1, \quad P(0) = 0.$$

Example 2 — Solution

Solution. Define $Q(x) = P(x) - x^2$. Then

$$\begin{aligned} Q(x+1) &= P(x+1) - (x+1)^2 = (P(x) + 2x + 1) - (x^2 + 2x + 1) = P(x) - x^2 \\ &= Q(x). \end{aligned}$$

So $Q(x+1) = Q(x)$ for every real x .

Iterating: $Q(0) = Q(1) = Q(2) = \cdots = Q(n)$ for every non-negative integer n .

Now consider $R(x) = Q(x) - Q(0)$. This polynomial has infinitely many roots (every non-negative integer), yet R is a polynomial. By the Identity Principle in its strongest form,

$$R(x) \equiv 0 \implies Q(x) \equiv Q(0).$$

Since $P(0) = 0$, we have $Q(0) = -0^2 + 0 = 0$, so $Q \equiv 0$ and

$$\boxed{P(x) = x^2.}$$



The Identity Principle — Strong Form

Strong Identity Principle.

A polynomial with infinitely many roots is identically zero. Equivalently, two polynomials that agree on an infinite set are equal.

Mental model. A non-zero polynomial of degree n has at most n roots — a *finite* number. So “infinitely many roots” forces the degree to be undefined (i.e. the zero polynomial).

Typical use cases.

- Functional equations for polynomials — transform to obtain $Q(x) \equiv 0$.
- Showing two polynomial expressions are equal by checking on $\mathbb{N}$ or any infinite set.
- Justifying “the formula derived for integers extends to all reals.”

The Rational Root Theorem

Rational Root Theorem.

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with *integer* coefficients, $a_n \neq 0$, $a_0 \neq 0$. If $\frac{p}{q}$ is a rational root in lowest terms (i.e. $\gcd(p, q) = 1$), then

$$p \mid a_0 \quad \text{and} \quad q \mid a_n.$$

Monic Corollary.

If $P(x)$ is *monic* with integer coefficients ($a_n = 1$), then every rational root is in fact an *integer dividing the constant term* a_0 .

A Crucial Caution

△ **Warning.** The Rational Root Theorem is **not** an existence statement. It does *not* say that a rational root must exist. It says:

IF a rational root p/q exists, *THEN* $p \mid a_0$ and $q \mid a_n$.

The polynomial $x^2 - 2$ has integer coefficients with $a_0 = -2$, $a_n = 1$. The candidates are $\pm 1, \pm 2$ — none of which is a root. The conclusion is not “no rational root exists by RRT” but rather “the only possible rational roots are these four candidates, and none works, hence no rational root exists.”

The right way to use RRT. Treat it as a *filter*: it produces a finite list of candidates. You then verify each. If none works, irrationality is established.

Example — $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is Irrational

Problem. Prove that $\beta = \sqrt{2} + \sqrt{3} + \sqrt{5}$ is irrational.

Why the Coefficients Are Integers

Solution. Define

$$P(x) = \prod_{\varepsilon \in \{\pm 1\}^3} (x - \varepsilon_1\sqrt{2} - \varepsilon_2\sqrt{3} - \varepsilon_3\sqrt{5}).$$

Pair factors that differ only in the sign of $\sqrt{5}$:

$$(x - a - \sqrt{5})(x - a + \sqrt{5}) = (x - a)^2 - 5,$$

where $a = \varepsilon_1\sqrt{2} + \varepsilon_2\sqrt{3}$. This eliminates $\sqrt{5}$, leaving four factors involving only $\sqrt{2}$ and $\sqrt{3}$.

Now pair those four by the sign of $\sqrt{3}$: each such pair eliminates $\sqrt{3}$ in the same way. We are left with two factors involving only $\sqrt{2}$.

Pair the last two by the sign of $\sqrt{2}$: this eliminates $\sqrt{2}$ as well.

At every stage the elementary identity $(u - v)(u + v) = u^2 - v^2$ removes one surd. After three rounds, all radicals are gone — $P(x)$ is a monic polynomial of degree 8 with *integer* coefficients.

Finishing the Argument

Solution (contd.) Setting all signs to $+1$ shows $P(\beta) = 0$ (one of the eight factors is exactly $x - \beta$). So β is a root of a monic integer polynomial.

Suppose $\beta \in \mathbb{Q}$. By the Monic Corollary, β must be an integer dividing the constant term $P(0)$.

But we can bound β tightly:

$$\sqrt{2} > 1.41, \quad \sqrt{3} > 1.73, \quad \sqrt{5} > 2.23 \quad \implies \quad \beta > 5.37,$$

$$\sqrt{2} < 1.42, \quad \sqrt{3} < 1.74, \quad \sqrt{5} < 2.24 \quad \implies \quad \beta < 5.40.$$

So $5 < \beta < 6$, and β is not an integer — contradiction. Hence β is irrational. ■

The general pattern. For any sum $\alpha = \sqrt{n_1} + \sqrt{n_2} + \cdots + \sqrt{n_k}$, take the product over all 2^k sign choices. Successive pairings eliminate one radical at a time, yielding a *monic* integer polynomial of degree 2^k with α as a root. The same Monic-Corollary-and-bound trick then establishes irrationality.

A Recurring Setup

In several competition problems, the polynomial in question is monic with non-negative real coefficients, has all real roots, and a fixed constant term. The structural deductions below will be used *repeatedly*.

Lemma. Let $P(x)$ be a monic polynomial of degree n with non-negative real coefficients. If P has n real roots, then *all roots are non-positive*. If additionally the constant term is positive, all roots are strictly negative.

Proof. For $x \geq 0$, every term of $P(x)$ is non-negative and the leading term $x^n > 0$ (for $x > 0$), so $P(x) > 0$. Hence no root lies in $(0, \infty)$. If $P(0) = a_0 > 0$, then 0 is not a root either, so all roots are strictly negative.

Writing the roots as $-r_1, \dots, -r_n$ with $r_i > 0$:

$$P(x) = \prod_{i=1}^n (x + r_i), \quad P(0) = \prod_{i=1}^n r_i.$$

Problem 3 — A Root-Bounding Inequality

Problem. Let $P(x)$ be a monic polynomial of degree n with non-negative real coefficients and constant term 1. If $P(x)$ has n real roots, prove that

$$P(2) \geq 3^n.$$

Problem 3 — Solution

Solution. By the lemma, the roots are $-r_1, \dots, -r_n$ with $r_i > 0$, and the constant term gives

$$P(0) = \prod_{i=1}^n r_i = 1.$$

Evaluating at $x = 2$:

$$P(2) = \prod_{i=1}^n (2 + r_i) = \prod_{i=1}^n (1 + 1 + r_i).$$

By AM–GM applied to the three non-negative numbers $1, 1, r_i$:

$$\frac{1 + 1 + r_i}{3} \geq \sqrt[3]{1 \cdot 1 \cdot r_i} \implies 2 + r_i \geq 3r_i^{1/3}.$$

Multiplying over $i = 1, \dots, n$:

$$P(2) \geq \prod_{i=1}^n 3r_i^{1/3} = 3^n \left(\prod_{i=1}^n r_i \right)^{1/3} = 3^n \cdot 1^{1/3} = 3^n. \quad \blacksquare$$

Problem 4 — Sum of Coefficients

Problem. Let $P(x)$ be a monic polynomial of degree n with non-negative real coefficients and constant term 1. If P has n real roots, prove that the sum of its coefficients is at least 2^n .

Crux. The sum of coefficients is $P(1)$. Apply AM–GM to the pair $(1, r_i)$ in each factor.

Problem 4 — Solution

Solution. The sum of the coefficients of P equals $P(1)$. With roots $-r_i$ ($r_i > 0$, $\prod r_i = 1$):

$$P(1) = \prod_{i=1}^n (1 + r_i).$$

By AM–GM on $(1, r_i)$:

$$1 + r_i \geq 2\sqrt{r_i}.$$

Multiplying:

$$P(1) \geq \prod_{i=1}^n 2\sqrt{r_i} = 2^n \sqrt{\prod_{i=1}^n r_i} = 2^n. \quad \blacksquare$$

Pattern emerging. For Huygens'-style problems, the trick is consistently: split $k + r_i$ favourably, apply AM–GM term-by-term, multiply, and use $\prod r_i = 1$.

Problem 5 — Bounding Individual Coefficients

Problem. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + 1$ have non-negative real coefficients and n real roots. Prove that for each $1 \leq k \leq n-1$,

$$a_k \geq \binom{n}{k}.$$

Crux. Vieta expresses a_k as a sum of $\binom{n}{k}$ symmetric products. Apply AM–GM to that sum and use the symmetry of multiplicities.

Problem 5 — Solution

Solution. Roots are $-r_1, \dots, -r_n$ with $r_i > 0$, $\prod r_i = 1$.

By Vieta's formulas, a_k equals the elementary symmetric polynomial $e_{n-k}(r_1, \dots, r_n)$, which is the sum of all $\binom{n}{n-k} = \binom{n}{k}$ products of $n - k$ distinct r_i 's:

$$a_k = \sum_{|S|=n-k} \prod_{i \in S} r_i.$$

Apply AM–GM to these $\binom{n}{k}$ non-negative summands:

$$\frac{a_k}{\binom{n}{k}} \geq \left(\prod_{|S|=n-k} \prod_{i \in S} r_i \right)^{1/\binom{n}{k}}.$$

Problem 5 — Solution (contd.)

Solution (contd.) In the giant product on the right, each individual r_i appears in exactly those subsets S that contain i , and there are $\binom{n-1}{n-k-1}$ such subsets. By symmetry, every r_i has the same multiplicity, so

$$\prod_{|S|=n-k} \prod_{i \in S} r_i = \left(\prod_{i=1}^n r_i \right)^{\binom{n-1}{n-k-1}} = 1^{\binom{n-1}{n-k-1}} = 1.$$

Therefore

$$\frac{a_k}{\binom{n}{k}} \geq 1^{1/\binom{n}{k}} = 1, \quad \text{i.e. } a_k \geq \binom{n}{k}. \quad \blacksquare$$

Verification. Setting $r_1 = \cdots = r_n = 1$ gives $P(x) = (x+1)^n$, whose coefficients are exactly $\binom{n}{k}$. Equality holds, confirming the bound is sharp.

Problem 6 — A Cauchy–Schwarz Extension

Problem. Let $P(x) = x^n - a_1x^{n-1} + a_2x^{n-2} - \cdots + (-1)^n a_n$ have n strictly positive real roots. Prove that

$$a_1 a_{n-1} \geq n^2 a_n.$$

Crux. Vieta with positive roots: a_1 is the sum of roots, a_n the product, and a_{n-1} relates to the sum of reciprocals. The inequality reduces directly to Cauchy–Schwarz on (r_i) and $(1/r_i)$.

Problem 6 — Solution

Solution. Let the positive roots be $r_1, \dots, r_n > 0$. By Vieta's formulas (note the alternating sign convention),

$$a_1 = \sum_{i=1}^n r_i, \quad a_n = \prod_{i=1}^n r_i, \quad a_{n-1} = \sum_{i=1}^n \prod_{j \neq i} r_j = a_n \sum_{i=1}^n \frac{1}{r_i}.$$

Dividing the desired inequality $a_1 a_{n-1} \geq n^2 a_n$ by $a_n > 0$:

$$a_1 \cdot \frac{a_{n-1}}{a_n} \geq n^2 \quad \Longleftrightarrow \quad \left(\sum_{i=1}^n r_i \right) \left(\sum_{i=1}^n \frac{1}{r_i} \right) \geq n^2.$$

By Cauchy–Schwarz (with $u_i = \sqrt{r_i}$, $v_i = 1/\sqrt{r_i}$),

$$\left(\sum r_i \right) \left(\sum \frac{1}{r_i} \right) \geq \left(\sum \sqrt{r_i} \cdot \frac{1}{\sqrt{r_i}} \right)^2 = n^2. \quad \blacksquare$$

Problem 7 — Huygens' Inequality for Polynomials

Problem. Let $P(x)$ be a monic polynomial of degree n with non-negative real coefficients, n real roots, and constant term 1. For every real $k > 0$, prove that

$$P(k) \geq (k + 1)^n.$$

Crux. A direct application of *Huygens' inequality* (the multiplicative form of AM–GM): for non-negative a_i, b_i ,

$$\prod_{i=1}^n (a_i + b_i) \geq \left((\prod a_i)^{1/n} + (\prod b_i)^{1/n} \right)^n.$$

Problem 7 — Solution

Solution. Roots are $-r_i$ ($r_i > 0$, $\prod r_i = 1$). Then

$$P(k) = \prod_{i=1}^n (k + r_i).$$

Apply Huygens' inequality with $a_i = k$, $b_i = r_i$:

$$\prod_{i=1}^n (k + r_i) \geq \left(\left(\prod_{i=1}^n k \right)^{1/n} + \left(\prod_{i=1}^n r_i \right)^{1/n} \right)^n = (k + 1^{1/n})^n = (k + 1)^n. \quad \blacksquare$$

Specializations recovered.

- $k = 1$: gives $P(1) \geq 2^n$ — this is Problem 4.
- $k = 2$: gives $P(2) \geq 3^n$ — this is Problem 3.

Huygens unifies the two earlier results into a single one-line argument.

Problem 8 — Schur's Theorem on Polynomials

Problem. Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that the set of primes p such that $p \mid P(n)$ for some integer n is *infinite*.

Crux. Assume only finitely many such primes $\{p_1, \dots, p_k\}$. Use a clever substitution that produces a polynomial value congruent to a fixed nonzero integer modulo each p_i , exposing a new prime divisor — contradiction.

Problem 8 — Solution

Solution. If $P(0) = 0$, write $P(x) = x^m Q(x)$ with $Q(0) \neq 0$; the result for Q implies it for P . So assume $c := P(0) \neq 0$.

Suppose, for contradiction, that the set of primes dividing some $P(n)$ is finite, say $\{p_1, \dots, p_k\}$. Let

$$M = c \cdot p_1 p_2 \cdots p_k.$$

Consider $P(xM)$. Writing $P(y) = c + a_1 y + a_2 y^2 + \cdots + a_n y^n$,

$$P(xM) = c + M \cdot (a_1 x + a_2 M x^2 + \cdots + a_n M^{n-1} x^n) = c(1 + M' \cdot R(x)),$$

where $M' = p_1 \cdots p_k$ and $R(x)$ is some integer polynomial (since $M = cM'$, every coefficient after c is divisible by cM' , so factoring out c leaves M' inside).

Problem 8 — Solution (contd.)

Solution (contd.) For sufficiently large $|x|$, the integer $N(x) := 1 + M'R(x)$ satisfies $|N(x)| > 1$, hence has some prime divisor q .

Claim: $q \notin \{p_1, \dots, p_k\}$.

Indeed, $N(x) \equiv 1 \pmod{p_i}$ for every i , so no p_i can divide $N(x)$.

Now $q \mid N(x)$ implies $q \mid c \cdot N(x) = P(xM)$. So q is a prime that divides some value of P , namely $P(xM)$. But q is not in the supposedly complete list $\{p_1, \dots, p_k\}$ — a contradiction.

Hence the set of primes dividing values of P is infinite. ■

Cultural note. This is Schur's analogue of Euclid's theorem on infinitely many primes: $P(x) = x$ recovers Euclid's classical argument almost verbatim.

Problem 9 — Parity in Polynomial Cycles

Problem. Let $P(x) \in \mathbb{Z}[x]$. Prove that there do *not* exist an odd integer $k \geq 3$ and distinct integers $x_1, x_2, \dots, x_k$ such that

$$P(x_1) = x_2, \quad P(x_2) = x_3, \quad \dots, \quad P(x_{k-1}) = x_k, \quad P(x_k) = x_1.$$

Crux. The fundamental lemma $a - b \mid P(a) - P(b)$ creates a cyclic chain of divisibilities. Mutual divisibility forces all consecutive differences to share absolute value, and a parity argument on the cyclic sum closes the case.

Problem 9 — Solution

Solution. Lemma. For any $P \in \mathbb{Z}[x]$ and integers a, b , $a - b \mid P(a) - P(b)$.
(Each $a^j - b^j$ is divisible by $a - b$, and $P(a) - P(b)$ is an integer combination of such terms.)

Apply the lemma to consecutive pairs in the cycle:

$$x_1 - x_2 \mid P(x_1) - P(x_2) = x_2 - x_3.$$

Iterating around the cycle yields the chain

$$x_1 - x_2 \mid x_2 - x_3 \mid x_3 - x_4 \mid \cdots \mid x_k - x_1 \mid x_1 - x_2.$$

Mutual divisibility of nonzero integers forces equality of absolute values:

$$|x_1 - x_2| = |x_2 - x_3| = \cdots = |x_k - x_1| =: d > 0.$$

(They are nonzero because the x_i are distinct.)

Problem 9 — Solution (contd.)

Solution (contd.) So each consecutive difference equals $\pm d$:

$$x_{i+1} - x_i \in \{+d, -d\} \quad \text{for } i = 1, \dots, k \text{ (indices mod } k).$$

Summing around the full cycle:

$$\sum_{i=1}^k (x_{i+1} - x_i) = 0,$$

by telescoping. Let p be the count of $+d$ and q the count of $-d$, so $p + q = k$. The cyclic sum equals $(p - q)d = 0$, forcing $p = q$ (since $d \neq 0$), and hence

$$k = p + q = 2p \quad \text{is even.}$$

This contradicts k odd. ■

Takeaway. Even cycles ($k = 2$, i.e. $P(P(x)) = x$ on integer pairs) are perfectly possible. The obstruction is genuinely an odd-cycle phenomenon.

Toolkit Summary — Part I: Identity & Roots

Strategic Patterns for ISI UGB Polynomial Problems.

- **Identity Principle:** *To prove $P \equiv Q$ for $\deg \leq n$, verify at $n + 1$ points; for any degree, exhibit infinitely many roots.*
- **Functional Recursions:** subtract a candidate solution to obtain Q with infinitely many roots, then $Q \equiv 0$.
- **Rational Root Theorem:** a candidate filter, not an existence claim. Combine with bounding for irrationality proofs.
- **Monic Corollary:** every rational root of a monic integer polynomial is an integer dividing the constant term.

Toolkit Summary — Part II: Inequalities & Number Theory

- **Vieta + AM–GM / Cauchy–Schwarz:** the standard route from coefficient inequalities to root inequalities and back.
- **Huygens' inequality:** unifies $P(k) \geq (k+1)^n$ -type bounds for monic non-negative-coefficient polynomials with $\prod r_i = 1$.
- **Divisibility lemma** $a - b \mid P(a) - P(b)$: the workhorse for integer-cycle and parity arguments.
- **Schur's substitution** $x \mapsto xM$: when finitely many primes are assumed, manufacture a polynomial value coprime to all of them.

Meta-principle. Most polynomial problems reduce to one of three moves: *exhibit enough roots* (Identity Principle), *exploit Vieta* (relate roots to coefficients), or *use modular structure* (divisibility, RRT, Schur).

End of Lecture
